Shaoyang Zhang

Computer Science · University of Michigan, Ann Arbor

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RESEARCH INTERESTS

Mechanistic interpretability, world-model safety, adversarial robustness, cognition-aware AI, and human–AI interaction.

EDUCATION

University of Michigan, Ann Arbor

B.S.E. in Computer Science | Aug 2025 – May 2027 (expected)

Cheng Family Scholarship (2025); Dean’s List.

Shanghai Jiao Tong University

B.E. in Mechanical Engineering | Aug 2023 – Aug 2027 (expected)

Academic Excellence Scholarship (2024). President, ASME UMJI Student Chapter; Vice President, Students Science & Technology Innovation Association.

RESEARCH EXPERIENCE

Northwestern University

Undergraduate Research Assistant · Remote | Mar 2026 – Present

Lead ongoing research on world-model hijacking and targeted adversarial attacks that aim to steer model behavior toward an attacker-chosen objective. Contributed controlled interventions and layer-wise representation analysis to research on attention sinks and activation outliers.

Michigan Institute for Data and AI in Society (MIDAS)

Undergraduate Research Assistant · Ann Arbor | Oct 2025 – Sep 2026

Mentors: Prof. Q. Vera Liao and Dr. Yoonjoo Lee. Built conversation-analysis pipelines and experimental infrastructure for knowledge-state-aware user simulation, including information-unit extraction, state updates, user-query generation, and evaluation. Implemented the initial end-to-end human validation-study system, including the chat interface, response collection, participant progress tracking, and data logging. Iterated through pilot feedback and failure analysis.

Data-informed Design and Intelligent Systems Laboratory

Undergraduate Research Assistant · Shanghai | Oct 2024 – Oct 2025

Mentors: Prof. Youyi Bi and Dr. Chang Liu. Studied imitation learning for robotic manipulation; reproduced and evaluated MimicPlay. Independently developed keyboard-based robotic-arm teleoperation to collect human demonstration trajectories.

Shaoyang Zhang

Publications & technical experience

PUBLICATIONS

Attention Sinks and Outliers in Attention Residuals

Haozheng Luo*, Haoran Dai*, Jingyuan Huang*, **Shaoyang Zhang**, Xi Chen, Eric Hanchen Jiang, Yijiang Li, Chenghao Qiu, Chenwei Xu, Zhenyu Pan, Haotian Zhang, Binghui Wang, and Yan Chen

NeurIPS 2026 · COLM 2026 Efficient Reasoning Workshop · Spotlight

https://arxiv.org/abs/2605.17887

KnowSim: Evaluating Information Calibration in LLM Assistants with User Simulators that Learn

Yoonjoo Lee, Hyoungwook Jin, Tae Soo Kim, **Shaoyang Zhang**, Philippe Laban, and Q. Vera Liao

EMNLP 2026 · Main Conference

https://aclanthology.org/2026.emnlp-main.1307/

Knowledge Graph Enhanced Interactive Fault Diagnosis O&M Approach Based on Large Models of Electrical Power

Xiaoyun Tian, Bin Liu, Jianzhong Yang, Bo Ai, Huailiang Zhao, and **Shaoyang Zhang**

PSGEC 2024 · Power System and Green Energy Conference

https://ieeexplore.ieee.org/document/10721152

* Equal contribution. OASIS author ordering follows the current public paper.

SELECTED ENGINEERING PROJECTS

Robotic-arm teleoperation

DIDIS Laboratory

Developed keyboard-based teleoperation and collected demonstration trajectories for imitation learning experiments.

DATA TAMER

EECS 448 team project · In development, Fall 2026

Exploring StarCraft II win-probability prediction and attribution using spatial sequences and structured game-state features.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript.
ML: PyTorch, TensorFlow, Hugging Face Transformers.
Research: LLM pipelines, user simulation, knowledge-state modeling, mechanistic interpretability, activation and representation analysis, adversarial evaluation.
Robotics: MuJoCo, RoboVerse, imitation learning, teleoperation.
Tools: Weights & Biases, Git, Linux, LaTeX, MATLAB.